**We would like to acknowledge Stanford University's CS231n on which we based the development of this project.**

In [1]:
```
import os
import sys
sys.path.insert(0, os.path.dirname(os.path.abspath('gan.py')))
print('cwd:', os.getcwd())
```

```
cwd: c:\Kiro Workspace\ucla-engr-c147b-2026-spring-project-1\GAN
```

## Enabling GPU

To enable GPU on your gcloud instance, make sure you selected a GPU machine type when creating your VM. You can verify the GPU is available by running `nvidia-smi` in the terminal.

# Generative Adversarial Networks (GANs)

In C147/C247, all the applications of neural networks that we have explored have been **discriminative models** that take an input and are trained to produce a labeled output. In this notebook, we will expand our repetoire, and build **generative models** using neural networks. Specifically, we will learn how to build models which generate novel images that resemble a set of training images.

## What is a GAN?

In 2014, Goodfellow et al. presented a method for training generative models called Generative Adversarial Networks (GANs for short). In a GAN, we build two different neural networks. Our first network is a traditional classification network, called the **discriminator**. We will train the discriminator to take images and classify them as being real (belonging to the training set) or fake (not present in the training set). Our other network, called the **generator**, will take random noise as input and transform it using a neural network to produce images. The goal of the generator is to fool the discriminator into thinking the images it produced are real.

We can think of this back and forth process of the generator ($G$) trying to fool the discriminator ($D$) and the discriminator trying to correctly classify real vs. fake as a minimax game:

$$\underset{G}{\text{minimize}}\ \underset{D}{\text{maximize}}\ \mathbb{E}_{x\sim p_{\text{data}}}\left[\log D(x)\right] + \mathbb{E}_{z\sim p(z)}\left[\log(1 - D(G(z)))\right]$$

where $z \sim p(z)$ are the random noise samples, $G(z)$ are the generated images using the neural network generator $G$, and $D$ is the output of the discriminator, specifying the

probability of an input being real. In Goodfellow et al., they analyze this minimax game and show how it relates to minimizing the Jensen-Shannon divergence between the training data distribution and the generated samples from $G$.

To optimize this minimax game, we will alternate between taking gradient *descent* steps on the objective for $G$ and gradient *ascent* steps on the objective for $D$:

1. update the **generator** ($G$) to minimize the probability of the **discriminator making the correct choice**.
2. update the **discriminator** ($D$) to maximize the probability of the **discriminator making the correct choice**.

While these updates are useful for analysis, they do not perform well in practice. Instead, we will use a different objective when we update the generator: maximize the probability of the **discriminator making the incorrect choice**. This small change helps to allevaiate problems with the generator gradient vanishing when the discriminator is confident. This is the standard update used in most GAN papers and was used in the original paper from Goodfellow et al..

In this assignment, we will alternate the following updates:

1. Update the generator ($G$) to maximize the probability of the discriminator making the incorrect choice on generated data:

$$\underset{G}{\text{maximize}}\ \mathbb{E}_{z\sim p(z)}\left[\log D(G(z))\right]$$

2. Update the discriminator ($D$), to maximize the probability of the discriminator making the correct choice on real and generated data:

$$\underset{D}{\text{maximize}}\ \mathbb{E}_{x\sim p_{\text{data}}}\left[\log D(x)\right] + \mathbb{E}_{z\sim p(z)}\left[\log(1 - D(G(z)))\right]$$

Here's an example of what your outputs from the 3 different models you're going to train should look like. Note that GANs are sometimes finicky, so your outputs might not look exactly like this. This is just meant to be a *rough* guideline of the kind of quality you can expect:

# Assignment Overview

All your code goes in `gan.py`. The notebook imports from it and runs checks — do not modify the notebook cells.

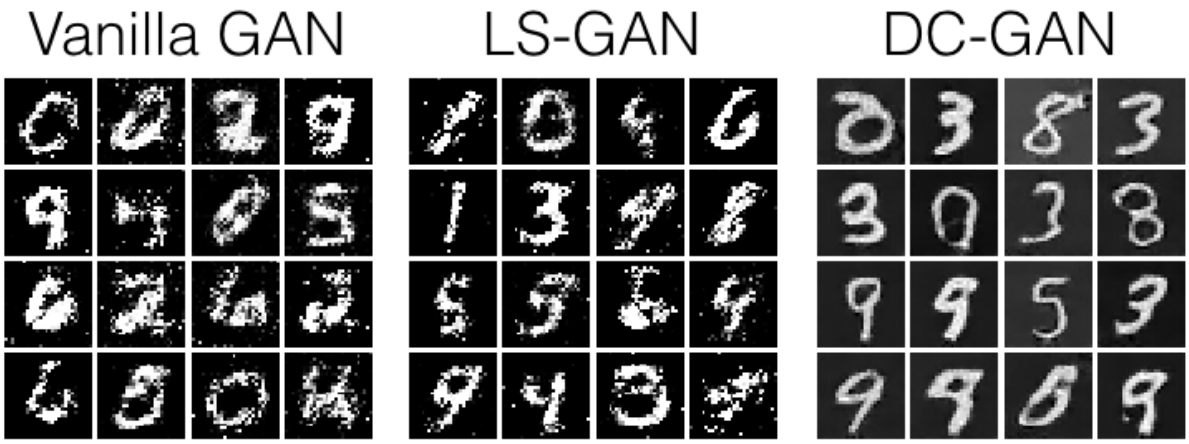
| Task | Points |
|---|---|
| `Discriminator` — MLP architecture | 1 |
| `Generator` — MLP architecture | 1 |

| Task | Points |
|---|---|
| `discriminator_loss` + `generator_loss` — BCE GAN loss | 2 |
| `ls_discriminator_loss` + `ls_generator_loss` — LSGAN loss | 2 |
| `DCDiscriminator` — convolutional architecture | 1 |
| `DCGenerator` — convolutional architecture | 1 |
| Inline Questions 4, 5, 6 | 3 |
| **Total** | **11** |

`sample_noise`, `get_optimizer`, `bce_loss`, and `run_a_gan` are already implemented for you.

In [2]:
```
# Run this cell to see sample outputs.
from IPython.display import Image
Image('assets/gan_outputs.png')
```

Out[2]:




In [3]:
```
# Setup cell.
import numpy as np
import torch
import torch.nn as nn
import torchvision.transforms as T
import torch.optim as optim
from torch.utils.data import DataLoader
import torchvision.datasets as dset
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from gan import rel_error, count_params

%matplotlib inline
plt.rcParams['figure.figsize'] = (10.0, 8.0)
plt.rcParams['image.interpolation'] = 'nearest'
plt.rcParams['image.cmap'] = 'gray'

%load_ext autoreload
%autoreload 2

def show_images(images):
```

```
    images = np.reshape(images, [images.shape[0], -1])
    sqrtn = int(np.ceil(np.sqrt(images.shape[0])))
    sqrtimg = int(np.ceil(np.sqrt(images.shape[1])))

    fig = plt.figure(figsize=(sqrtn, sqrtn))
    gs = gridspec.GridSpec(sqrtn, sqrtn)
    gs.update(wspace=0.05, hspace=0.05)

    for i, img in enumerate(images):
        ax = plt.subplot(gs[i])
        plt.axis('off')
        ax.set_xticklabels([])
        ax.set_yticklabels([])
        ax.set_aspect('equal')
        plt.imshow(img.reshape([sqrtimg, sqrtimg]))
    return

answers = dict(np.load('assets/gan-checks.npz'))
device = torch.device('cuda') if torch.cuda.is_available() else (torch.devic
print(device)
```

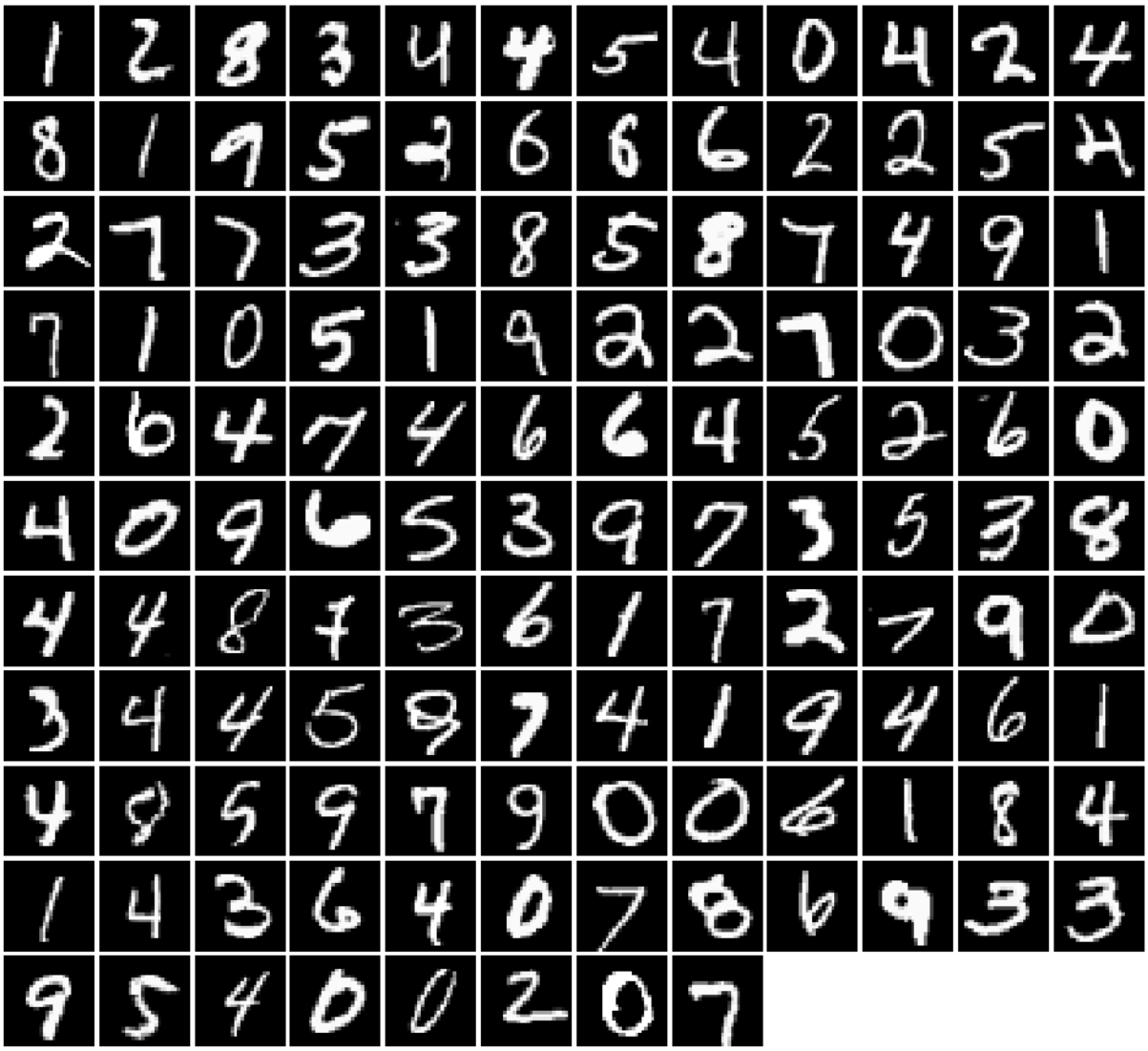
```
cuda
```

## Dataset

GANs are notoriously finicky with hyperparameters, and also require many training epochs. In order to make this assignment approachable, we will be working on the MNIST dataset, which is 60,000 training and 10,000 test images. Each picture contains a centered image of white digit on black background (0 through 9). This was one of the first datasets used to train convolutional neural networks and it is fairly easy -- a standard CNN model can easily exceed 99% accuracy.

To simplify our code here, we will use the PyTorch MNIST wrapper, which downloads and loads the MNIST dataset. See the documentation for more information about the interface. The data will be saved into a folder called `MNIST`.

In [4]:
```
NOISE_DIM = 96
batch_size = 128

mnist_train = dset.MNIST('./mnist_data', train=True, download=True, transfor
loader_train = DataLoader(mnist_train, batch_size=batch_size, shuffle=True,

iterator = iter(loader_train)
imgs, labels = next(iterator)
imgs = imgs.view(batch_size, 784).numpy().squeeze()
show_images(imgs)
```



## PyTorch Utilities

For the convolutional generator you will need to reshape tensors between flat vectors and image tensors. Use PyTorch's built-in `nn.Flatten()` and `nn.Unflatten(dim, shape)` — no need to import anything extra.

We also provide a weight initializer (and call it for you) that uses Xavier initialization instead of PyTorch's uniform default.

In [5]:
```
from gan import initialize_weights
```

# Discriminator (1 point)

Our first step is to build a discriminator. Fill in `self.model` as an `nn.Sequential` inside the `Discriminator` class in `gan.py`. All fully connected layers should include bias terms. The architecture is:

- Fully connected layer with input size 784 and output size 256
- LeakyReLU with alpha 0.01
- Fully connected layer with input_size 256 and output size 256
- LeakyReLU with alpha 0.01
- Fully connected layer with input size 256 and output size 1

Recall that the Leaky ReLU nonlinearity computes $f(x) = \max(\alpha x, x)$ for some fixed constant $\alpha$; for the LeakyReLU nonlinearities in the architecture above we set $\alpha = 0.01$.

The output of the discriminator should have shape `[batch_size, 1]`, and contain real numbers corresponding to the scores that each of the `batch_size` inputs is a real image.

Implement `Discriminator` in `gan.py`

Test to make sure the number of parameters in the discriminator is correct:

In [6]:
```
from gan import Discriminator

def test_discriminator(true_count=267009):
    model = Discriminator()
    cur_count = count_params(model)
    if cur_count != true_count:
        print('Incorrect number of parameters in discriminator. Check your a
    else:
        print('Correct number of parameters in discriminator.')

test_discriminator()
```

```
Correct number of parameters in discriminator.
```

# Generator (1 point)

Now to build the generator network. Fill in `self.model` as an `nn.Sequential` inside the `Generator` class in `gan.py`:

- Fully connected layer from noise_dim to 1024
- `ReLU`
- Fully connected layer with size 1024
- `ReLU`
- Fully connected layer with size 784
- `TanH` (to clip the image to be in the range of [-1,1])

All fully connected layers should include bias terms. Implement `Generator` in `gan.py`

Test to make sure the number of parameters in the generator is correct:

In [7]:
```
from gan import Generator

def test_generator(true_count=1858320):
    model = Generator(noise_dim=4)
    cur_count = count_params(model)
    if cur_count != true_count:
        print('Incorrect number of parameters in generator. Check your archi
    else:
        print('Correct number of parameters in generator.')

test_generator()
```

```
Correct number of parameters in generator.
```

# GAN Loss (2 points)

Compute the generator and discriminator loss. The generator loss is:

$$\ell_G = -\mathbb{E}_{z \sim p(z)} \left[ \log D(G(z)) \right]$$

and the discriminator loss is:

$$\ell_D = -\mathbb{E}_{x \sim p_{\text{data}}} \left[ \log D(x) \right] - \mathbb{E}_{z \sim p(z)} \left[ \log (1 - D(G(z))) \right]$$

Note that these are negated from the equations presented earlier as we will be *minimizing* these losses.

**HINTS**: You should use the `bce_loss` function defined below to compute the binary cross entropy loss which is needed to compute the log probability of the true label given the logits output from the discriminator. Given a score $s \in \mathbb{R}$ and a label $y \in \{0, 1\}$, the binary cross entropy loss is

$$bce(s, y) = -y * \log(s) - (1 - y) * \log(1 - s)$$

A naive implementation of this formula can be numerically unstable, so we have provided a numerically stable implementation that relies on PyTorch's `nn.BCEWithLogitsLoss`.

You will also need to compute labels corresponding to real or fake and use the logit arguments to determine their size. Make sure you use the `torch.float32` data type (same as `torch.float`) for compatibility with BCEWithLogitsLoss, which uses floating point data types, and move tensors to GPU (if applicable) using `u = u.to(device)`.

`true_labels = torch.ones(size, dtype=torch.float32).to(device)`

or

`true_labels = torch.ones(size, device=device) # uses torch.float32 by default`

Instead of computing the expectation of $\log D(G(z))$, $\log D(x)$ and $\log(1 - D(G(z)))$, we will be averaging over elements of the minibatch. This is taken care of in `bce_loss` which combines the loss by averaging. **Hint**: Do NOT concatenate real and fake losses. Sum the two (empirical) expectations instead.

Implement `discriminator_loss` and `generator_loss` in `gan.py`

Test your generator and discriminator loss. You should see errors < 1e-7.

In [8]:
```
from gan import bce_loss, discriminator_loss, generator_loss

def test_discriminator_loss(logits_real, logits_fake, d_loss_true):
    d_loss = discriminator_loss(
        torch.tensor(logits_real, dtype=torch.float32).to(device),
        torch.tensor(logits_fake, dtype=torch.float32).to(device)
    ).cpu().numpy()
    print(f'Maximum error in d_loss: {rel_error(d_loss_true, d_loss):g}')

test_discriminator_loss(
    answers['logits_real'],
    answers['logits_fake'],
    answers['d_loss_true']
)
```

```
Maximum error in d_loss: 3.97058e-09
```

In [9]:
```
def test_generator_loss(logits_fake, g_loss_true):
    g_loss = generator_loss(
        torch.tensor(logits_fake, dtype=torch.float32).to(device)
    ).cpu().numpy()
    print(f'Maximum error in g_loss: {rel_error(g_loss_true, g_loss):g}')

test_generator_loss(
    answers['logits_fake'],
    answers['g_loss_true']
)
```

```
Maximum error in g_loss: 3.4188e-08
```

# Training a GAN!

We provide you the main training loop. You won't need to change `run_a_gan` in `gan.py`, but we encourage you to read through it for your own understanding. Training takes about 7 minutes on CPU and about 1-2 minutes on GPU.

If it doesn't work the first time, check the discriminator: Did you use `nn.Flatten()`?

In [10]:
```
from gan import get_optimizer, run_a_gan

D = Discriminator().to(device)
G = Generator().to(device)
```

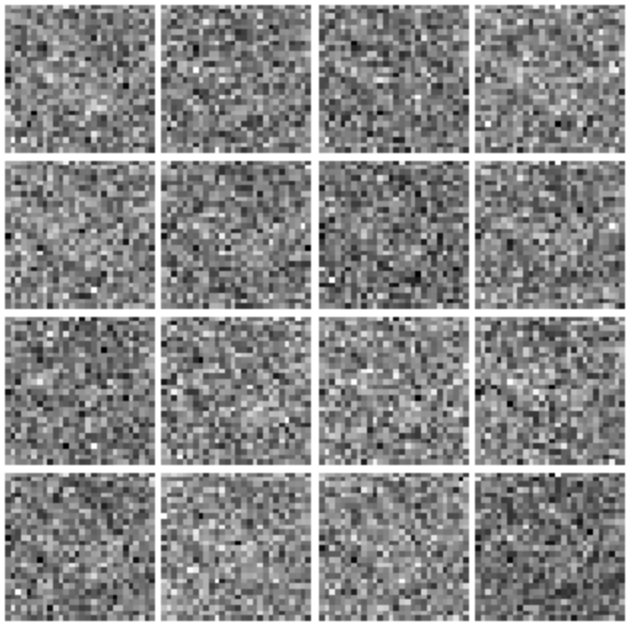
```
D_solver = get_optimizer(D)
G_solver = get_optimizer(G)

images = run_a_gan(
    D,
    G,
    D_solver,
    G_solver,
    discriminator_loss,
    generator_loss,
    loader_train
)
```

```
Iter: 0, D: 1.4246, G: 0.6939
Iter: 250, D: 1.5922, G: 0.9484
Iter: 500, D: 1.1297, G: 0.7062
Iter: 750, D: 1.0762, G: 1.1219
Iter: 1000, D: 1.2247, G: 1.1089
Iter: 1250, D: 1.2243, G: 0.9659
Iter: 1500, D: 1.2359, G: 0.8103
Iter: 1750, D: 1.3137, G: 0.8911
Iter: 2000, D: 1.2626, G: 0.9961
Iter: 2250, D: 1.2353, G: 1.0397
Iter: 2500, D: 1.3113, G: 0.8019
Iter: 2750, D: 1.3126, G: 0.9474
Iter: 3000, D: 1.3376, G: 0.8792
Iter: 3250, D: 1.2976, G: 0.7730
Iter: 3500, D: 1.2954, G: 0.7812
Iter: 3750, D: 1.2955, G: 0.8594
Iter: 4000, D: 1.3220, G: 0.7503
Iter: 4250, D: 1.3281, G: 0.7838
Iter: 4500, D: 1.3530, G: 0.8237
```

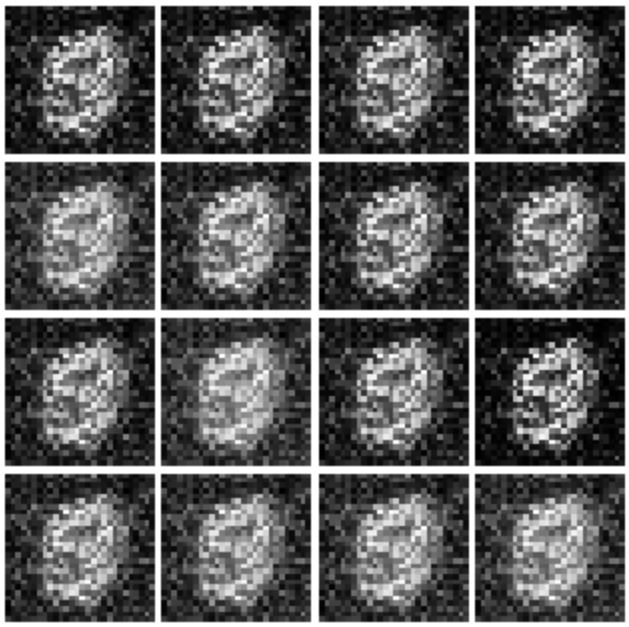
Run the cell below to show the generated images.

In [11]:
```
numIter = 0
for img in images:
    print(f'Iter: {numIter}')
    show_images(img)
    plt.show()
    numIter += 250
    print()
```

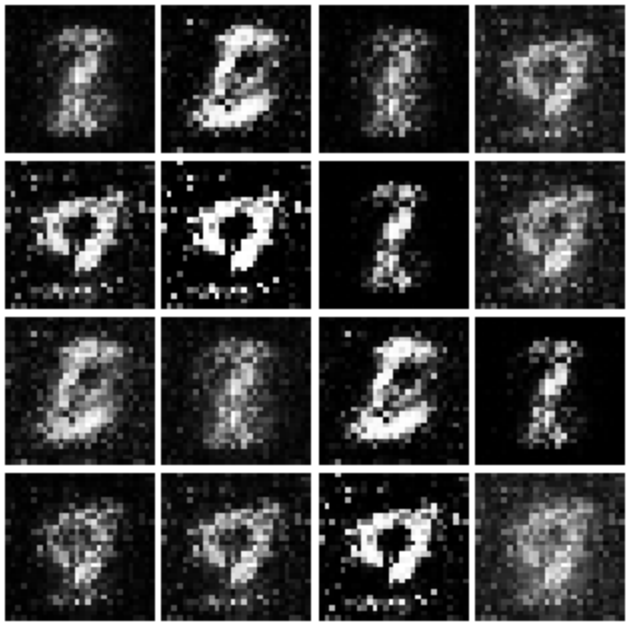
```
Iter: 0
```



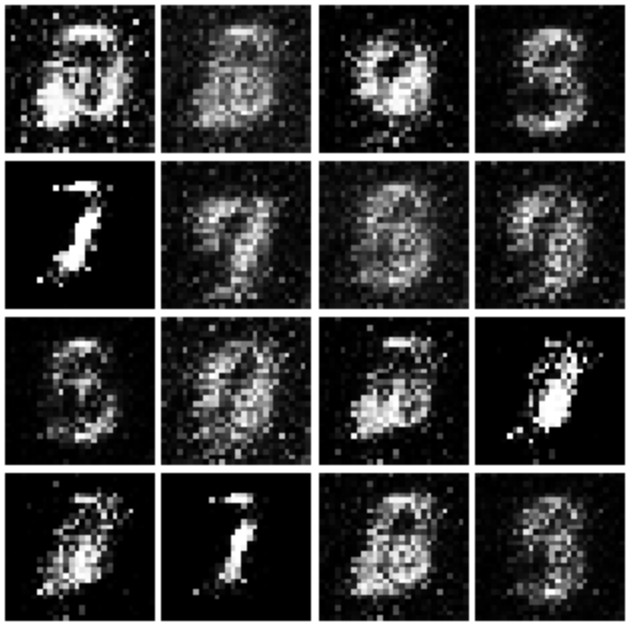
Iter: 250



Iter: 500

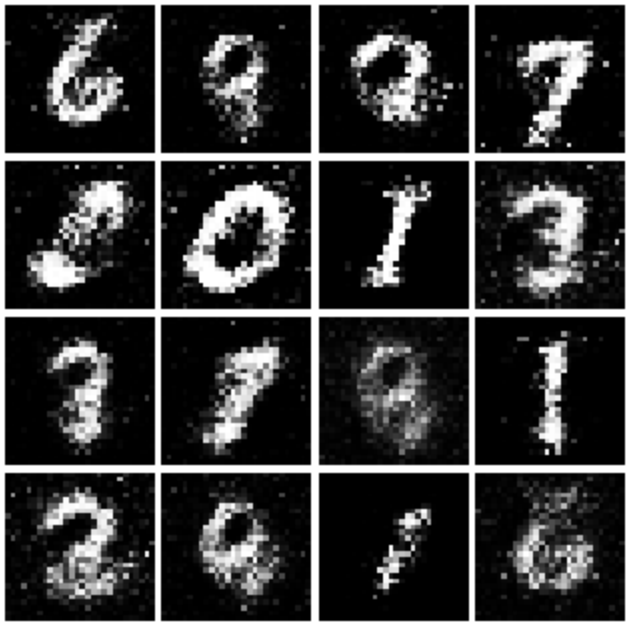


Iter: 750

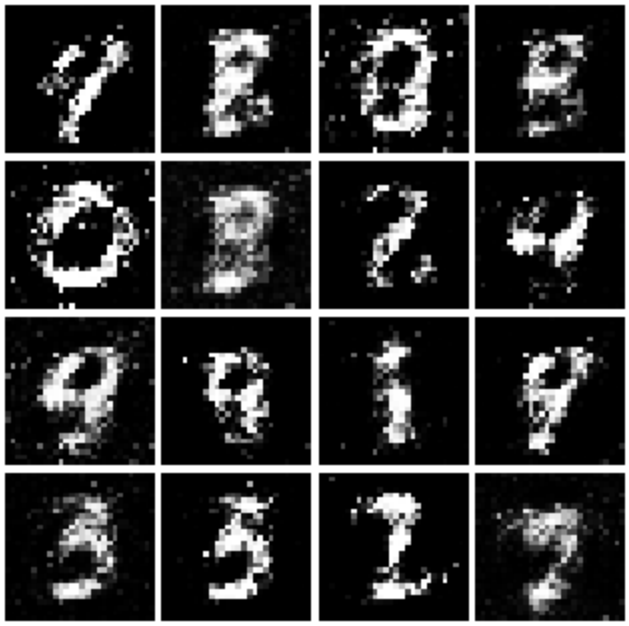


Iter: 1000

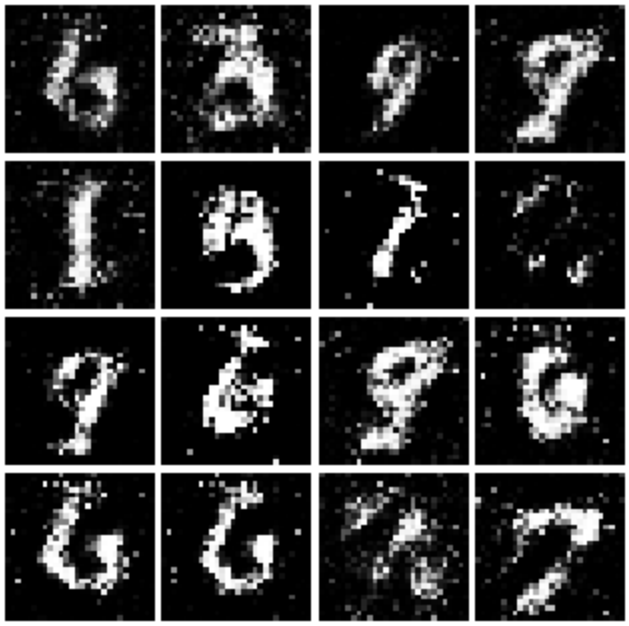
Iter: 1250



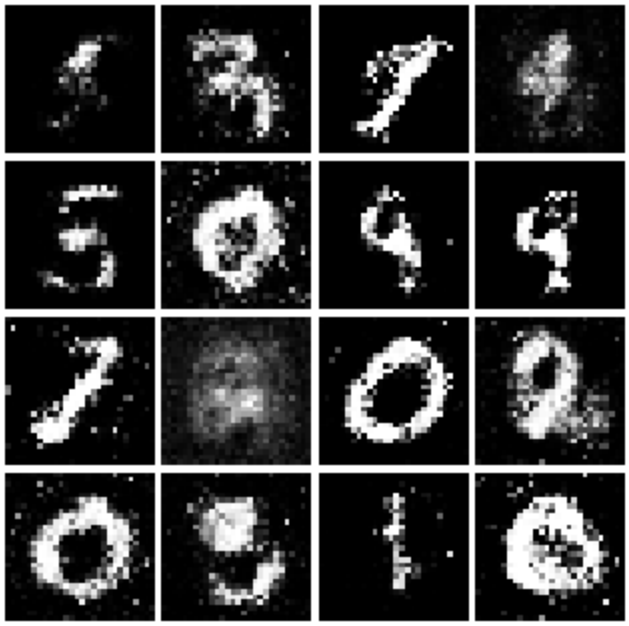
Iter: 1500



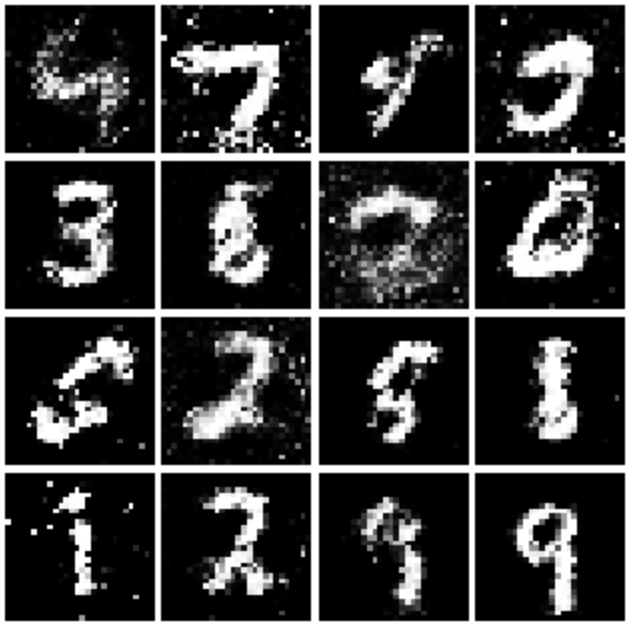
Iter: 1750



Iter: 2000

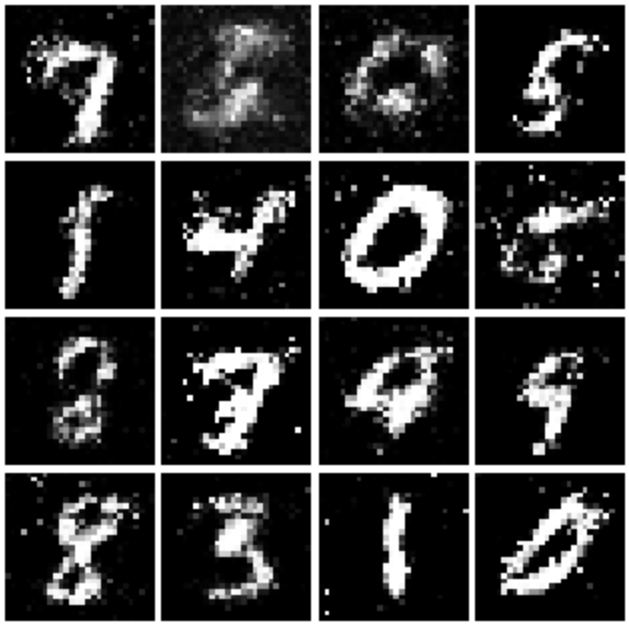


Iter: 2250

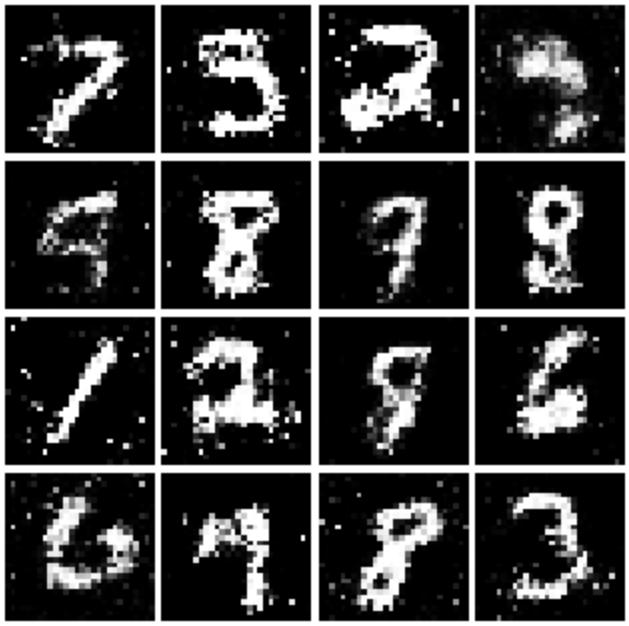


Iter: 2500

Iter: 2750

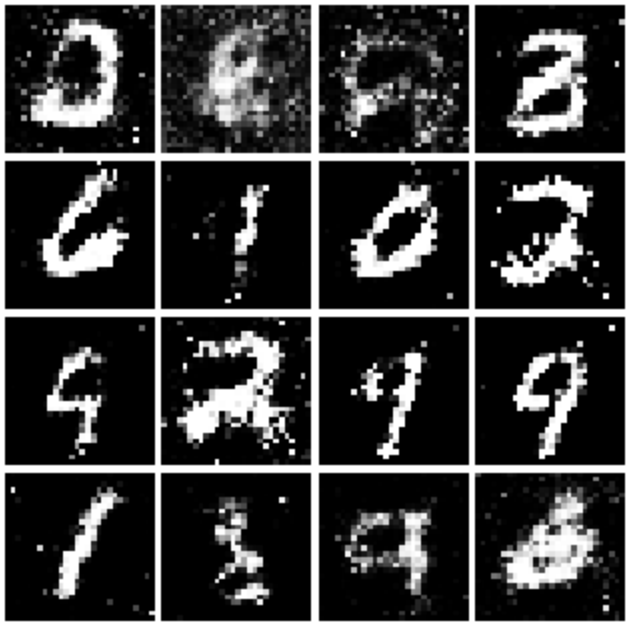


Iter: 3000



Iter: 3250

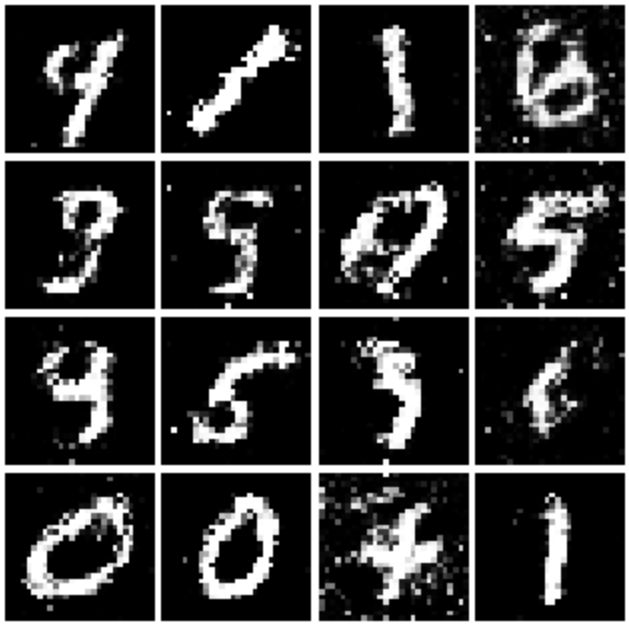
Iter: 3500



Iter: 3750

Iter: 4000

Iter: 4250



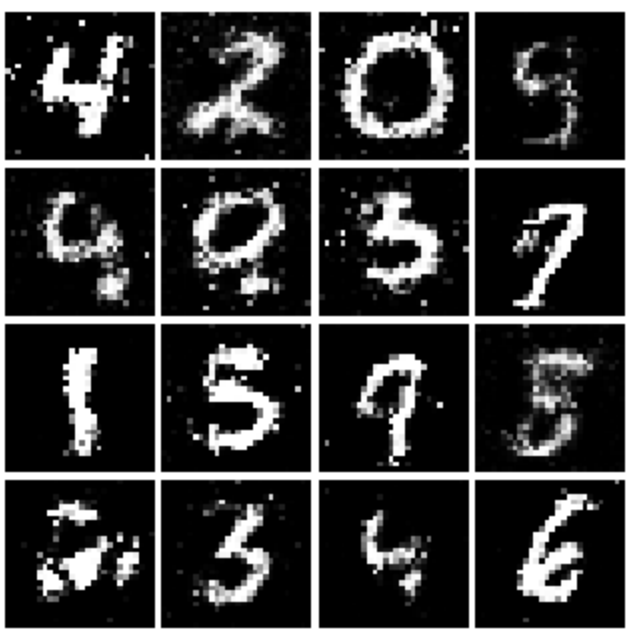
Iter: 4500

## Inline Question 1

What does your final vanilla GAN image look like?

In [12]:
```
# This output is your answer.
print('Vanilla GAN final image:')
show_images(images[-1])
plt.show()
```

```
Vanilla GAN final image:
```



Well that wasn't so hard, was it? In the iterations in the low 100s you should see black backgrounds, fuzzy shapes as you approach iteration 1000, and decent shapes, about half of which will be sharp and clearly recognizable as we pass 3000.

# Least Squares GAN (2 points)

We'll now look at Least Squares GAN, a newer, more stable alernative to the original GAN loss function. For this part, all we have to do is change the loss function and retrain the model. We'll implement equation (9) in the paper, with the generator loss:

$$\ell_G = \frac{1}{2}\mathbb{E}_{z \sim p(z)}\left[\left(D(G(z)) - 1\right)^2\right]$$

and the discriminator loss:

$$\ell_D = \frac{1}{2}\mathbb{E}_{x \sim p_{\text{data}}}\left[\left(D(x) - 1\right)^2\right] + \frac{1}{2}\mathbb{E}_{z \sim p(z)}\left[\left(D(G(z))\right)^2\right]$$

**HINTS**: Instead of computing the expectation, we will be averaging over elements of the minibatch, so make sure to combine the loss by averaging instead of summing. When plugging in for $D(x)$ and $D(G(z))$ use the direct output from the discriminator ( `scores_real` and `scores_fake` ). **If** you get errors of 0.333..., you probably forgot to include the 1/2 term.

Implement `ls_discriminator_loss` , `ls_generator_loss` in `gan.py`

Before running a GAN with our new loss function, let's check it:

In [13]:
```
from gan import ls_discriminator_loss, ls_generator_loss

def test_lsgan_loss(score_real, score_fake, d_loss_true, g_loss_true):
    score_real = torch.tensor(score_real, dtype=torch.float32).to(device)
    score_fake = torch.tensor(score_fake, dtype=torch.float32).to(device)
    d_loss = ls_discriminator_loss(score_real, score_fake).cpu().numpy()
    g_loss = ls_generator_loss(score_fake).cpu().numpy()
    print(f'Maximum error in d_loss: {rel_error(d_loss_true, d_loss):g}')
    print(f'Maximum error in g_loss: {rel_error(g_loss_true, g_loss):g}')

test_lsgan_loss(
    answers['logits_real'],
    answers['logits_fake'],
    answers['d_loss_lsgan_true'],
    answers['g_loss_lsgan_true']
)
```

```
Maximum error in d_loss: 1.53171e-08
Maximum error in g_loss: 2.7837e-09
```

Run the following cell to train your model! Training takes about 7 minutes on CPU and about 1-2 minutes on GPU.

In [14]:
```
D_LS = Discriminator().to(device)
G_LS = Generator().to(device)

D_LS_solver = get_optimizer(D_LS)
G_LS_solver = get_optimizer(G_LS)

images = run_a_gan(
    D_LS,
    G_LS,
    D_LS_solver,
    G_LS_solver,
    ls_discriminator_loss,
    ls_generator_loss,
    loader_train
)
```

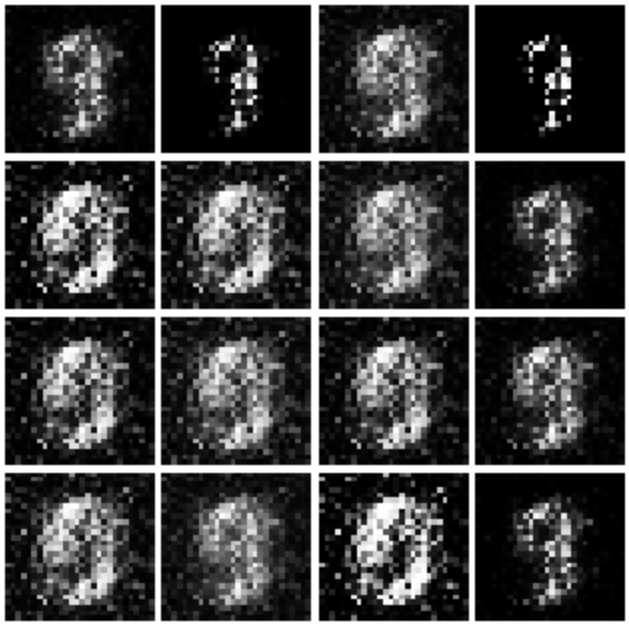
```
Iter: 0, D: 0.4792, G: 0.4660
Iter: 250, D: 0.1558, G: 0.3555
Iter: 500, D: 0.0894, G: 0.5689
Iter: 750, D: 0.2144, G: 0.2357
Iter: 1000, D: 0.2719, G: 0.2967
Iter: 1250, D: 0.1942, G: 0.4808
Iter: 1500, D: 0.2219, G: 0.2012
Iter: 1750, D: 0.2072, G: 0.2495
Iter: 2000, D: 0.2154, G: 0.1820
Iter: 2250, D: 0.2429, G: 0.1549
Iter: 2500, D: 0.2325, G: 0.1658
Iter: 2750, D: 0.2301, G: 0.1549
Iter: 3000, D: 0.2113, G: 0.1650
Iter: 3250, D: 0.2086, G: 0.1591
Iter: 3500, D: 0.2119, G: 0.1552
Iter: 3750, D: 0.2366, G: 0.1607
Iter: 4000, D: 0.2216, G: 0.1824
Iter: 4250, D: 0.2216, G: 0.1501
Iter: 4500, D: 0.2269, G: 0.1537
```

Run the cell below to show generated images.

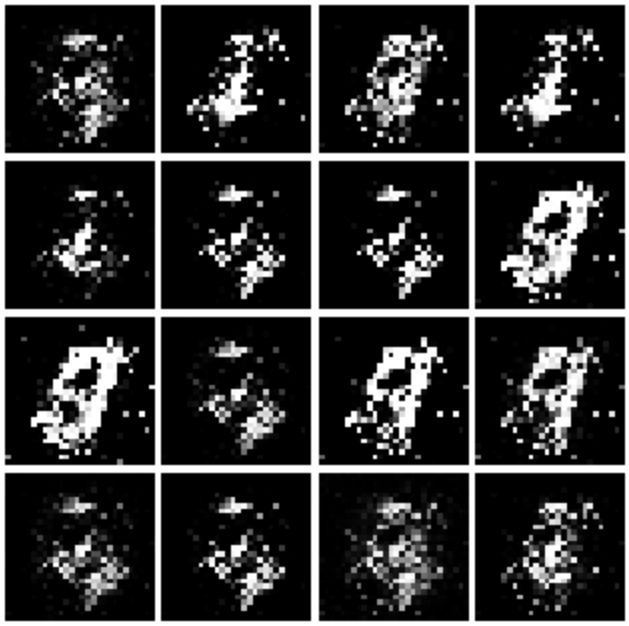
In [15]:
```
numIter = 0
for img in images:
    print(f'Iter: {numIter}')
    show_images(img)
    plt.show()
    numIter += 250
    print()
```

```
Iter: 0
```

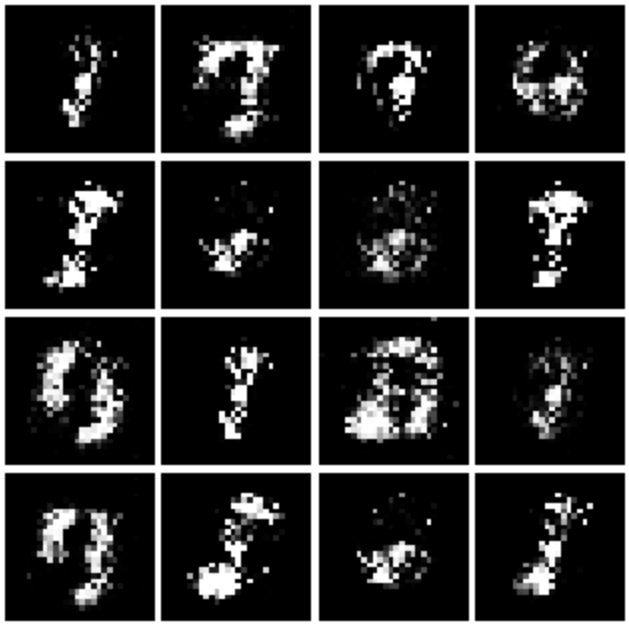
Iter: 250



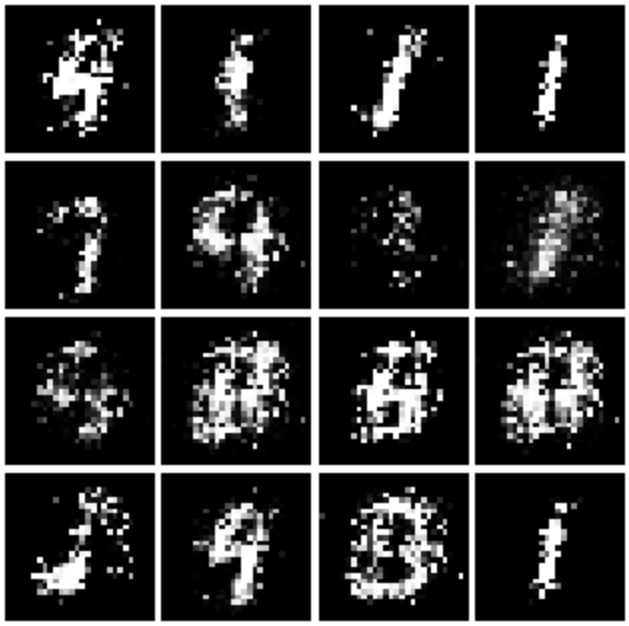
Iter: 500



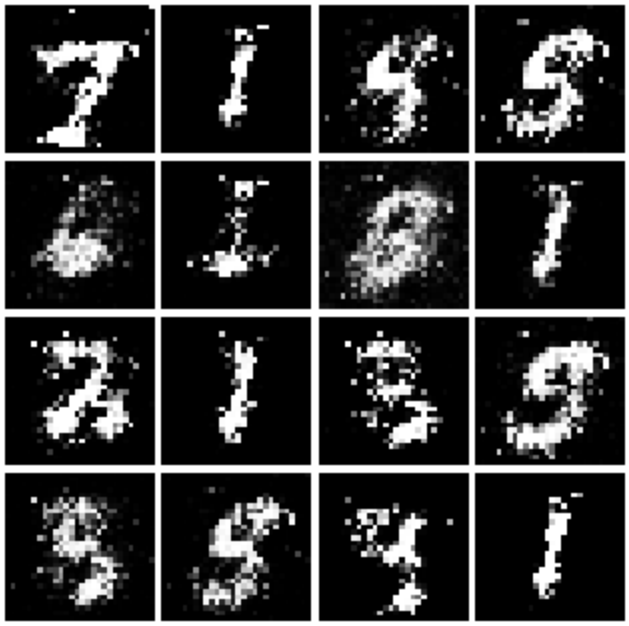
Iter: 750



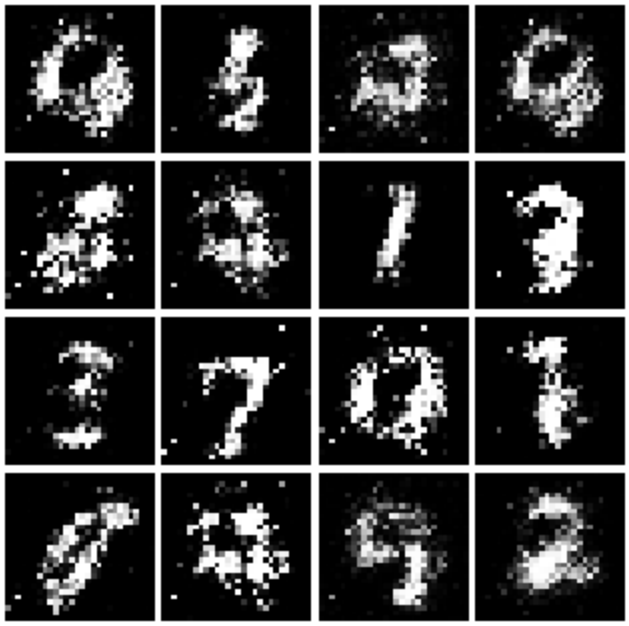
Iter: 1000



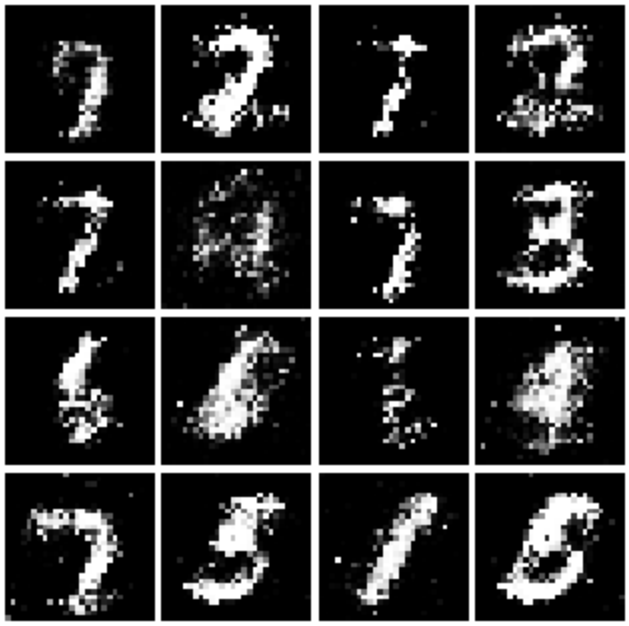
Iter: 1250



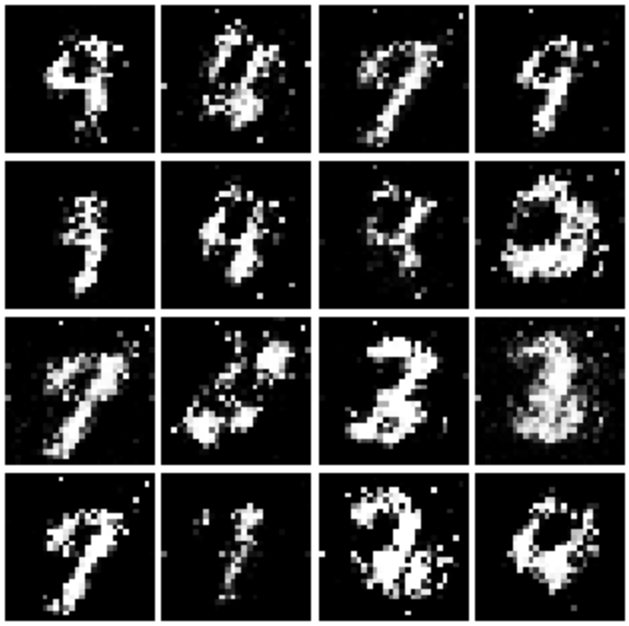
Iter: 1500



Iter: 1750



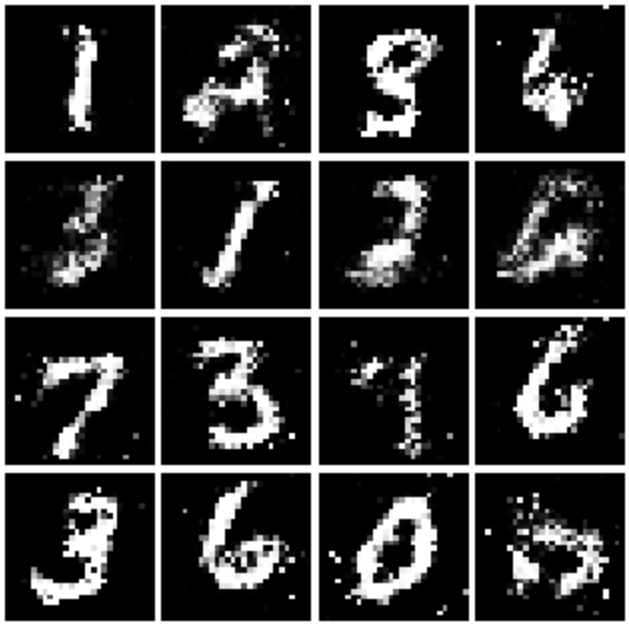
Iter: 2000



Iter: 2250

Iter: 2500

Iter: 2750



Iter: 3000

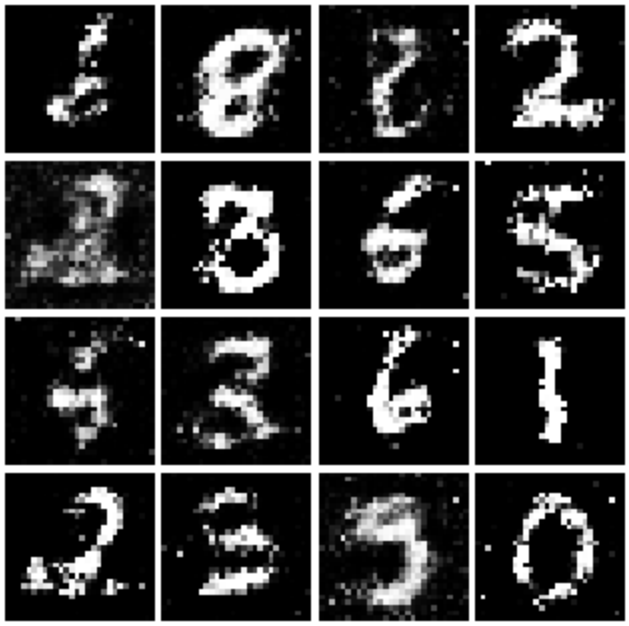
Iter: 3250

Iter: 3500

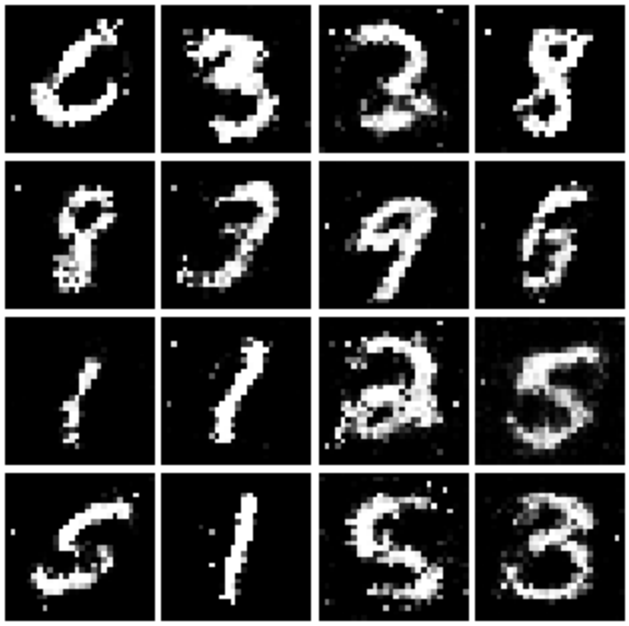
Iter: 3750

Iter: 4000

Iter: 4250



Iter: 4500

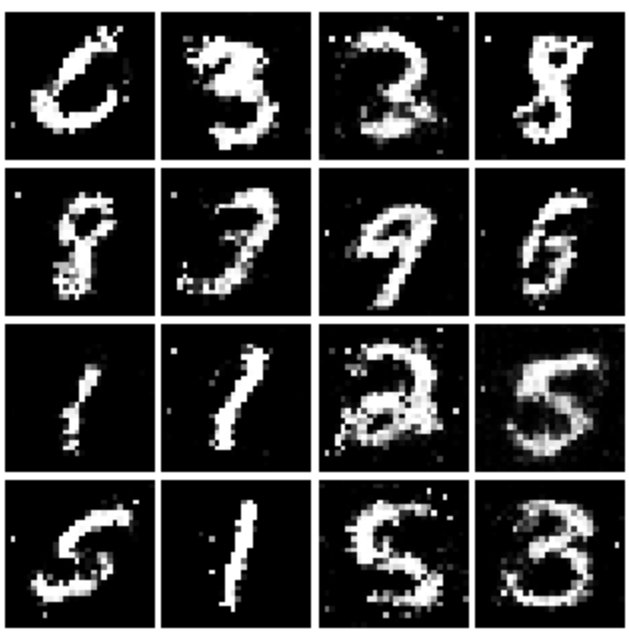


## Inline Question 2

What does your final LSGAN image look like?

In [16]:
```
# This output is your answer.
print('LSGAN final image:')
show_images(images[-1])
plt.show()
```

```
LSGAN final image:
```



# Deep Convolutional GAN (2 points)

In the first part of the notebook, we implemented an almost direct copy of the original GAN network from Ian Goodfellow. However, this network architecture allows no real spatial reasoning. It is unable to reason about things like "sharp edges" in general because it lacks any convolutional layers. Thus, in this section, we will implement some of the ideas from DCGAN, where we use convolutional networks

### Discriminator

We will use a discriminator inspired by the TensorFlow MNIST classification tutorial, which is able to get above 99% accuracy on the MNIST dataset fairly quickly. **Hint**: There is no need to specify padding. **Warning**: If you use LazyLinear I will take off points.

- Conv2D: 32 Filters, 5x5, Stride 1
- Leaky ReLU(alpha=0.01)
- Max Pool 2x2, Stride 2
- Conv2D: 64 Filters, 5x5, Stride 1
- Leaky ReLU(alpha=0.01)
- Max Pool 2x2, Stride 2
- Flatten
- Fully Connected with output size 4 x 4 x 64
- Leaky ReLU(alpha=0.01)
- Fully Connected with output size 1

Implement `DCDiscriminator` in `gan.py`

In [17]:
```
from gan import DCDiscriminator

data = next(enumerate(loader_train))[-1][0].to(device)
b = DCDiscriminator().to(device)
out = b(data)
print(out.size())
```

```
torch.Size([128, 1])
```

Check the number of parameters in your classifier as a sanity check:

In [18]:
```
def test_dc_classifer(true_count=1102721):
    model = DCDiscriminator()
    cur_count = count_params(model)
    if cur_count != true_count:
        print('Incorrect number of parameters in classifier. Check your arch
    else:
        print('Correct number of parameters in classifier.')

test_dc_classifer()
```

```
Correct number of parameters in classifier.
```

### Generator

For the generator, we will copy the architecture exactly from the InfoGAN paper. See Appendix C.1 MNIST. See the documentation for nn.ConvTranspose2d. We are always "training" in GAN mode.

- Fully connected with output size 1024
- `ReLU`
- BatchNorm
- Fully connected with output size 7 x 7 x 128
- ReLU
- BatchNorm
- Use `nn.Unflatten(1, (128, 7, 7))` to reshape into Image Tensor of shape 128, 7, 7
- `ConvTranspose2d`: 64 filters of 4x4, stride 2, 'same' padding (use `padding=1`)
- `ReLU`
- BatchNorm
- `ConvTranspose2d`: 1 filter of 4x4, stride 2, 'same' padding (use `padding=1`)
- `TanH`
- Should have a 28x28x1 image, reshape back into 784 vector (using `nn.Flatten()`)

Implement `DCGenerator` in `gan.py`

In [19]:
```
from gan import DCGenerator

test_g_gan = DCGenerator().to(device)
test_g_gan.apply(initialize_weights)

fake_seed = torch.randn(batch_size, NOISE_DIM).to(device)
fake_images = test_g_gan.forward(fake_seed)
fake_images.size()
```

Out[19]:
```
torch.Size([128, 784])
```

Check the number of parameters in your generator as a sanity check:

In [20]:
```
def test_dc_generator(true_count=6580801):
    model = DCGenerator(noise_dim=4)
    cur_count = count_params(model)
    if cur_count != true_count:
        print('Incorrect number of parameters in generator. Check your archi
    else:
        print('Correct number of parameters in generator.')

test_dc_generator()
```

```
Correct number of parameters in generator.
```

Run the following cell to train your DCGAN. Training takes about 35 minutes on CPU and about 1 minute on GPU.

Note: DCGAN uses the original BCE loss ( `discriminator_loss` / `generator_loss` ), not LSGAN. The architectural improvements in DCGAN (convolutional layers, batch norm) are what stabilize training here — LSGAN is a separate axis of improvement that you could combine with DCGAN, but for this assignment we keep them separate.

In [21]:
```
D_DC = DCDiscriminator().to(device)
D_DC.apply(initialize_weights)
G_DC = DCGenerator().to(device)
G_DC.apply(initialize_weights)

D_DC_solver = get_optimizer(D_DC)
G_DC_solver = get_optimizer(G_DC)

images = run_a_gan(
    D_DC,
    G_DC,
    D_DC_solver,
    G_DC_solver,
    discriminator_loss,
    generator_loss,
    loader_train,
    num_epochs=5
)
```

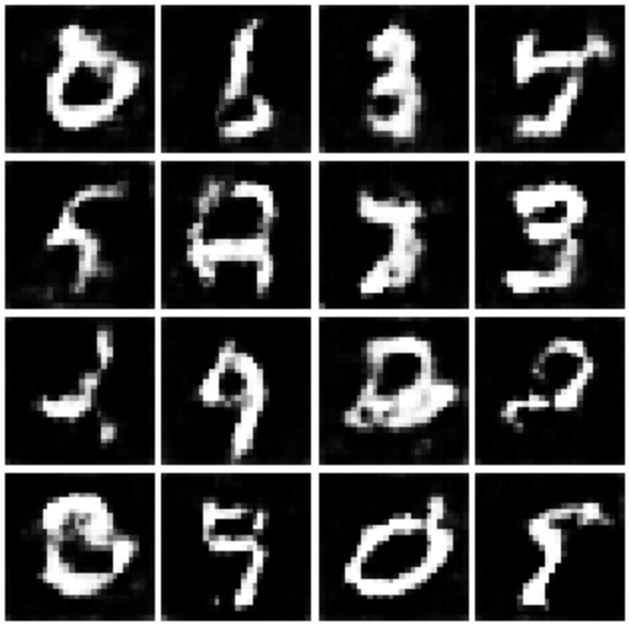
```
Iter: 0, D: 1.4907, G: 1.7457
Iter: 250, D: 1.0510, G: 0.7465
Iter: 500, D: 1.0695, G: 1.2029
Iter: 750, D: 1.1021, G: 1.0537
Iter: 1000, D: 1.0793, G: 1.0096
Iter: 1250, D: 1.1554, G: 1.0047
Iter: 1500, D: 1.1168, G: 1.3952
Iter: 1750, D: 1.1747, G: 1.1227
Iter: 2000, D: 1.1091, G: 1.0046
Iter: 2250, D: 1.0649, G: 1.1916
```

Run the cell below to show generated images.

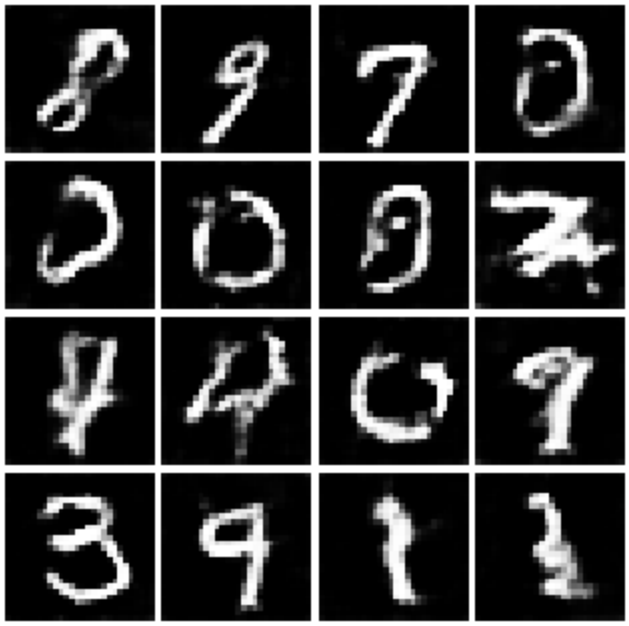
In [22]:
```
numIter = 0
for img in images:
    print(f'Iter: {numIter}')
    show_images(img)
    plt.show()
    numIter += 250
    print()
```

```
Iter: 0
```

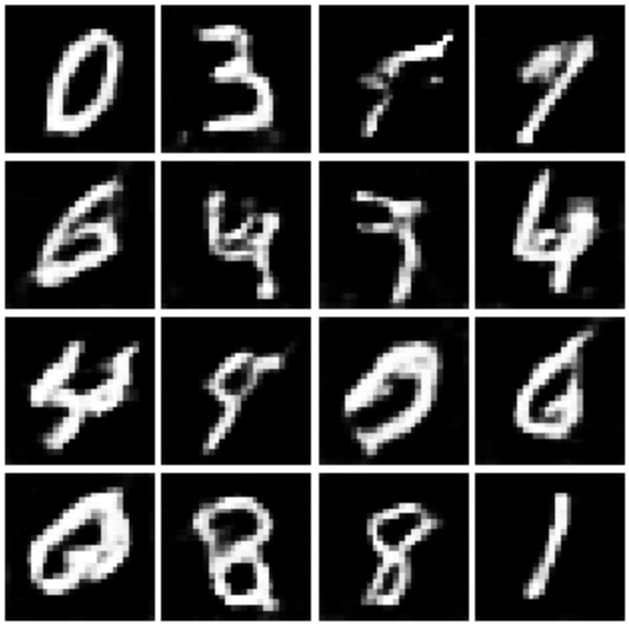
Iter: 250



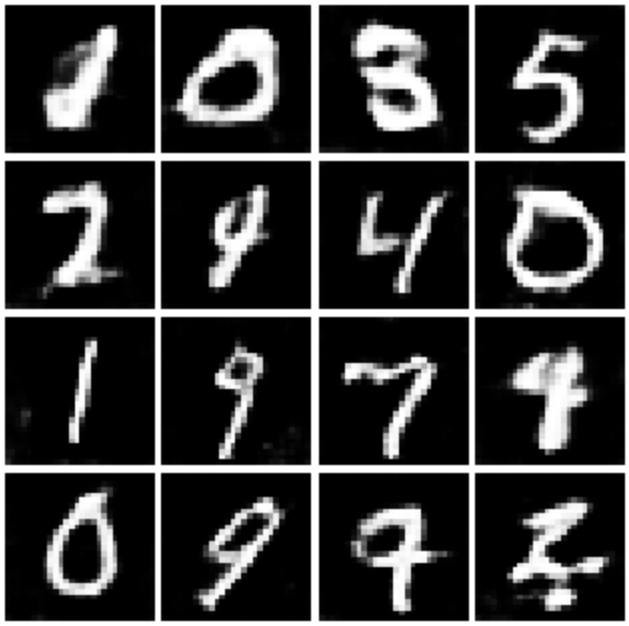
Iter: 500



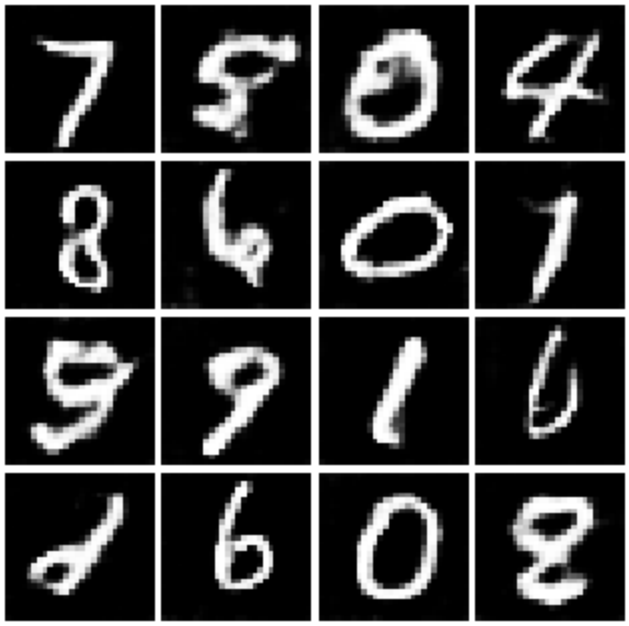
Iter: 750



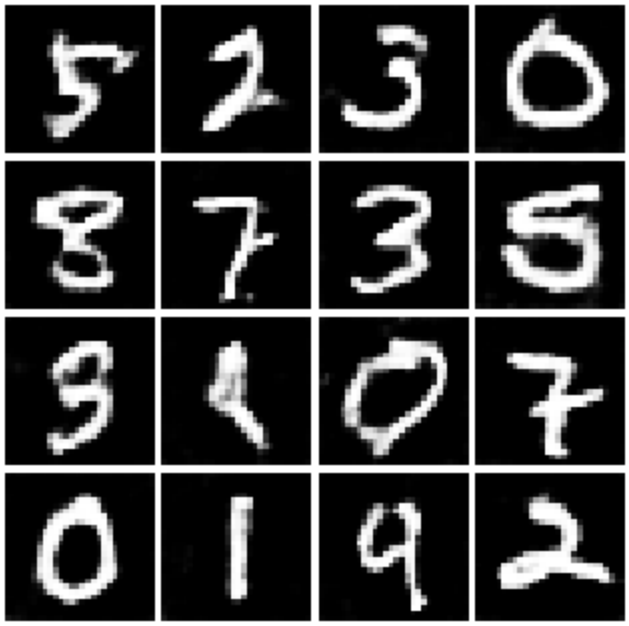
Iter: 1000



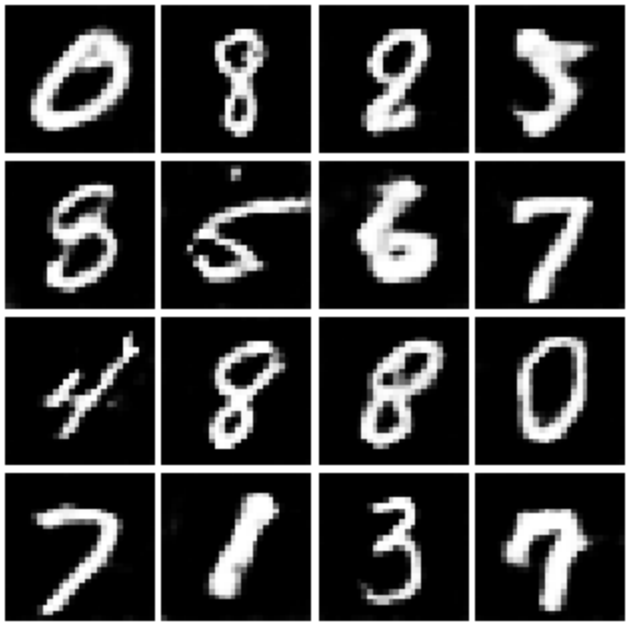
Iter: 1250



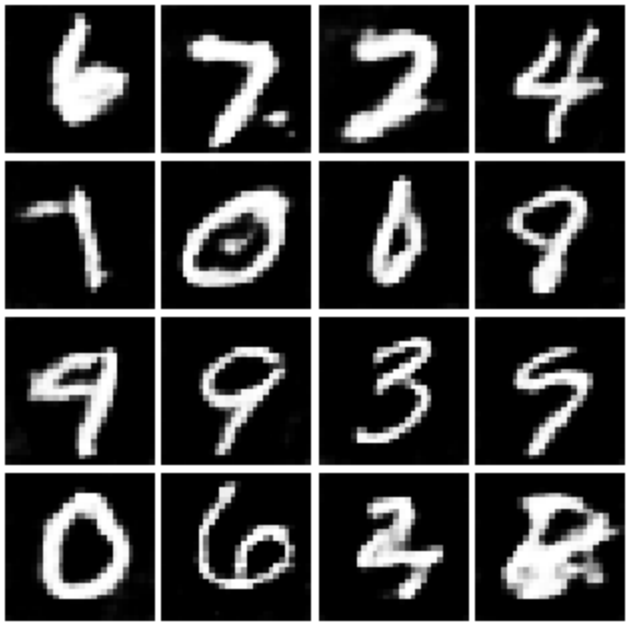
Iter: 1500



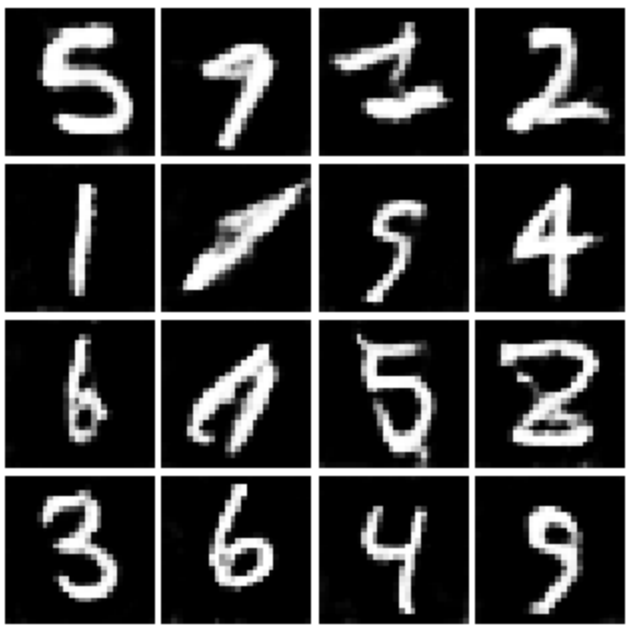
Iter: 1750



Iter: 2000



```
Iter: 2250
```

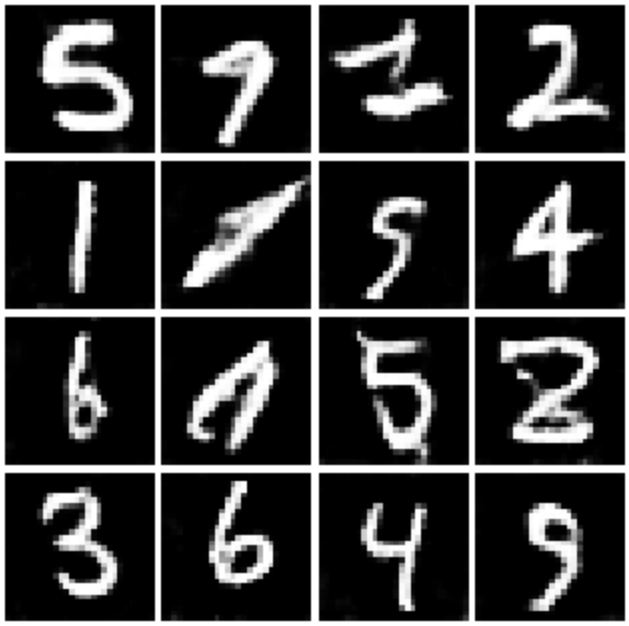


## Inline Question 3

What does your final DCGAN image look like?

In [23]:
```
# This output is your answer.
print('DCGAN final image:')
show_images(images[-1])
plt.show()
```

```
DCGAN final image:
```



## Inline Question 4 (1 point)

We will look at an example to see why alternating minimization of the same objective (like in a GAN) can be tricky business.

Consider $f(x, y) = xy$. What does $\min_x \max_y f(x, y)$ evaluate to? (0.4 points). Hint: minmax tries to minimize the maximum value achievable. Hint: Consider 3 separate cases: $x > 0$, $x = 0$, and $x < 0$.

Now try to evaluate this function numerically for 6 steps, starting at the point $(1, 1)$, by using alternating gradient (first updating y, then updating x using that updated y) with step size $1$. **Here step size is the learning_rate, and steps will be learning_rate * gradient.** You'll find that writing out the update step in terms of $x_t, y_t, x_{t+1}, y_{t+1}$ will be useful.

Record the six pairs of explicit values for $(x_t, y_t)$ in the table below and briefly explain what $f(x, y)$ evaluates to (0.6 points).

i.e., for $\eta = 1$, calculate:

For $i = 1, 2, 3, \ldots$

1. $y_i = y_{i-1} + \eta \frac{\partial f(x,y)}{\partial y}|_{x=x_{i-1}, y=y_{i-1}}$
2. $x_i = x_{i-1} - \eta \frac{\partial f(x,y)}{\partial x}|_{x=x_{i-1}, y=y_i}$

## Your answer:

**Part 1:** $\min_x \max_y f(x, y) = 0$.

Consider three cases:

- If $x > 0$: $\max_y xy = +\infty$ (send $y \to +\infty$).
- If $x = 0$: $\max_y xy = 0$ for all $y$.
- If $x < 0$: $\max_y xy = +\infty$ (send $y \to -\infty$).

So $\max_y xy$ is $+\infty$ unless $x = 0$, where it equals $0$. Therefore $\min_x \max_y xy = 0$, achieved at $x = 0$.

**Part 2:** With $f(x, y) = xy$, we have $\frac{\partial f}{\partial y} = x$ and $\frac{\partial f}{\partial x} = y$. The update rules with $\eta = 1$ are:

1. $y_i = y_{i-1} + x_{i-1}$ (gradient ascent on $y$)
2. $x_i = x_{i-1} - y_i$ (gradient descent on $x$, using the **updated** $y_i$)

| $y_0$ | $y_1$ | $y_2$ | $y_3$ | $y_4$ | $y_5$ | $y_6$ |
|---|---|---|---|---|---|---|
| 1 | 2 | 1 | -1 | -2 | -1 | 1 |
| $x_0$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
| 1 | -1 | -2 | -1 | 1 | 2 | 1 |

The values of $f(x, y) = xy$ at each step are: $f_0 = 1$, $f_1 = -2$, $f_2 = -2$, $f_3 = 1$, $f_4 = -2$, $f_5 = -2$, $f_6 = 1$. The iterates cycle with period 6 and never converge. The function oscillates between $1$ and $-2$, never reaching the optimal value of $0$.

# Inline Question 5 (1 point)

Using this method, will we ever reach the optimal value? Why or why not?

## Your answer:

No, we will never reach the optimal value of $0$. As shown in Question 4, the iterates cycle with period 6: $(x_6, y_6) = (x_0, y_0) = (1, 1)$. The alternating gradient method produces a fixed orbit that repeats indefinitely, so the values of $(x, y)$ never approach $(0, y)$ where the optimum is achieved. This happens because each player's gradient update overshoots and pushes the other player's variable in the opposite direction, creating a perpetual oscillation around the saddle point rather than converging to it. A smaller learning rate would reduce the magnitude of the oscillations but would not eliminate them — the iterates would still orbit the optimum without converging.

# Inline Question 6 (1 point)

If the generator loss decreases during training while the discriminator loss stays at a constant high value from the start, is this a good sign? Why or why not? A qualitative

answer is sufficient.

## Your answer:

No, this is not a good sign. A discriminator loss that stays at a constant high value from the start means the discriminator never learned to distinguish real images from fake ones — it is essentially guessing randomly the entire time. If the generator loss decreases against such a weak discriminator, it simply means the generator found a way to produce outputs that fool a clueless discriminator, not that it is generating realistic images. In a healthy GAN training, the discriminator and generator losses should both fluctuate as the two networks compete: the discriminator should improve at detecting fakes, which in turn forces the generator to produce more realistic images. A constantly poor discriminator provides no useful gradient signal, so the generator has no incentive to actually improve image quality.